

Deep Learning for NLP

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1. Abstract

This report studies the third QEvasion response-clarity assignment, where the supervised classifier from the previous two homeworks is replaced by prompt-based prediction with Qwen-family language models. The task remains unchanged: given a political interview question and answer, predict Clear Reply, Ambivalent, or Clear Non-Reply. The new system evaluates zero-shot prompting, few-shot prompting, a chain-of-thought-style instruction, and a self-check instruction across 0.8B, 2B, and 4B Qwen models.

The final notebook and script select the best prompting system by **validation macro F1**, with accuracy used only as a tie-breaker. This is intentional. Accuracy can reward systems that avoid the difficult middle class, while macro F1 exposes whether the model can detect Ambivalent. The best prompting run is `qwen-4b_few-shot`, with 0.4667 accuracy and 0.4667 macro F1 on a balanced 30-example validation sample. The most important finding is negative: small prompted language models repeatedly fail to identify Ambivalent. The 2B few-shot run has the highest raw accuracy at 0.5000, but it predicts zero Ambivalent examples, so it is not selected by the macro-F1 rule. Compared with the earlier classical and fine-tuned transformer systems, prompting is operationally convenient but substantially weaker on the core semantic boundary of the task.

2. Task and Prior Homeworks

2.1. Dataset

All three homeworks use the QEvasion dataset [1]. Each example contains a political interview question, the interviewee response, and a clarity label. The three target labels are:

- **Clear Reply**: the answer explicitly addresses the question;
- **Ambivalent**: the answer partly addresses the question, shifts scope, hedges, or gives only a partial response;
- **Clear Non-Reply**: the answer clearly does not answer, asks for clarification, claims ignorance, or declines.

The label taxonomy exposed by the dataset was included in the prompt as minimal task context:

Clear Reply ← Explicit,
Ambivalent ← Implicit, Dodging, General, Deflection, Partial,
Clear Non-Reply ← Declining, Ignorance, Clarification.

This is not extra supervision beyond the dataset definition; it is the natural label specification for the prompting task.

2.2. Continuity With Homeworks 1 and 2

Homework 1 solved the task with classical machine-learning representations.

The strongest system was TF-IDF plus Logistic Regression, with a reported test macro F1 of approximately 0.45 and accuracy around 0.63. Averaged GloVe representations performed worse, with macro F1 around 0.39 for Wiki/Gigaword vectors and 0.38 for Twitter vectors. The main lesson was that sparse lexical features preserve phrase-level cues that mean-pooled embeddings wash out.

Homework 2 replaced the classical encoder with fine-tuned transformer sequence classifiers. BERT-base achieved 0.6299 accuracy and 0.5561 macro F1 on the official test split. DistilBERT reached 0.5779 accuracy and 0.5066 macro F1. DeBERTa-v3 produced high accuracy but only 0.2672 macro F1 because the run collapsed to predicting Ambivalent for every test instance. That failure already showed why macro F1 is the right metric for this assignment: a system can look acceptable by accuracy and still be useless for two of the classes.

Homework 3 keeps the same prediction target but changes the learning regime completely. The Qwen models are not fine-tuned. They are used through generation, and the generated text is parsed back into one of the three classification labels.

3. Prompting System

3.1. Prompt Variants

The final pipeline runs four prompt strategies:

- **Zero-shot:** the model sees only the label list, the dataset taxonomy, the question, and the answer.
- **Few-shot:** the model also sees one balanced example per class sampled from the training split.
- **Chain-of-thought-style:** the model is encouraged to consider the responsiveness relation before giving a final label. The parser still uses only the final label.
- **Self-check:** the model is asked to verify that its final label is one of the allowed labels.

All prompt variants strongly constrain the output format. The generated answer must contain exactly one final label from Clear Reply, Ambivalent, and Clear Non-Reply. This was added because early runs showed that unconstrained generations were too verbose and hard to parse reliably.

3.2. Parsing and Evaluation Framework

Prompting does not directly produce class IDs, so the toolset includes a parser that maps generations back to dataset labels. The parser normalizes casing, hyphenation, and simple aliases, then validates that the output contains exactly one recognized clarity label. Invalid generations are tracked explicitly rather than silently coerced.

In the final run, output formatting was not the problem: every model/strategy combination had `valid_rate = 1.0` and `invalid_rate = 0.0`. The failures discussed below are therefore semantic failures, not parsing failures.

3.3. Execution Protocol

The final diagnostic run is stored under `runs_hw3_semifinal`. It uses:

- Qwen3.5 0.8B, 2B, and 4B models [6];
- the four prompt strategies above;
- deterministic decoding with temperature 0.0;
- `max_new_tokens = 48`;
- a balanced validation sample with 10 examples per label;
- macro F1 as the primary selection metric.

The balanced validation design is important. A 30-example sample is small, but it makes the failure mode visible: each class contributes exactly one third of the validation set, so a model cannot hide by following the natural class imbalance. Runtime is dominated by generation rather than training. The full validation grid requires $3 \times 4 \times 30 = 360$ generations, followed by one 308-example final submission pass for the selected system.

The final Kaggle notebook is [sdi2200160 homework-3-prompting](#). It writes the required file:

`submission_best_prompting_system.csv`.

4. Homework 3 Results

4.1. Validation Grid

Table 1 reports all final prompting runs. The table is sorted by macro F1, matching the notebook and script selection rule.

Model	Prompt	Acc.	Macro F1	Macro P	Macro R	Pred. CR	Pred. Amb.	Pred. CNR
4B	few-shot	0.4667	0.4667	0.4667	0.4667	10	10	10
4B	self-check	0.4667	0.4626	0.4639	0.4667	12	8	10
4B	zero-shot	0.4667	0.4626	0.5102	0.4667	9	16	5
0.8B	few-shot	0.4667	0.4091	0.6860	0.4667	23	1	6
2B	few-shot	0.5000	0.4044	0.4167	0.5000	24	0	6
4B	cot	0.4333	0.4002	0.3872	0.4333	13	5	12
0.8B	self-check	0.4333	0.3845	0.4841	0.4333	21	2	7
0.8B	cot	0.4667	0.3779	0.3685	0.4667	23	0	7
2B	zero-shot	0.4667	0.3683	0.4000	0.4667	25	0	5
2B	cot	0.4333	0.3333	0.3833	0.4333	25	1	4
2B	self-check	0.4333	0.3280	0.3782	0.4333	26	0	4
0.8B	zero-shot	0.3667	0.2315	0.4483	0.3667	29	0	1

Table 1: Final HW3 validation runs on a balanced 30-example sample. CR = Clear Reply; Amb. = Ambivalent; CNR = Clear Non-Reply. Selection is by macro F1, with accuracy as tie-breaker.

The selected system is `qwen-4b_few-shot`. It is not selected because it has the highest raw accuracy; it is selected because it is the best balanced classifier under macro F1.

The contrast with `qwen-2b_few-shot` is the key example: the 2B model reaches 0.5000 accuracy, but never predicts Ambivalent. That is exactly the kind of run the selection rule must reject.

4.2. Prompt and Model Trends

Few-shot prompting is the most reliable prompt family in this run. Averaged across model sizes, it gives the highest mean accuracy and mean macro F1. Self-check does not help; for the smaller models it mostly reinforces overconfident majority-like predictions. Chain-of-thought-style prompting is also not a clear win. The best chain-of-thought run, 4B CoT, is below the 4B zero-shot, self-check, and few-shot runs.

Model size matters mainly for Ambivalent. The 0.8B and 2B systems often collapse toward Clear Reply, with a small number of Clear Non-Reply predictions. The 4B model is the first one that consistently uses all three labels. Even then, its performance is weak: the best 4B run finds only 4 of the 10 Ambivalent validation examples.

4.3. Ambivalent Failure

Table 2 compares the selected 4B few-shot system against the accuracy-best 2B few-shot system.

System	True label	Pred. CR	Pred. Amb.	Pred. CNR
4B few-shot	Clear Reply	5	3	2
4B few-shot	Ambivalent	3	4	3
4B few-shot	Clear Non-Reply	2	3	5
2B few-shot	Clear Reply	10	0	0
2B few-shot	Ambivalent	9	0	1
2B few-shot	Clear Non-Reply	5	0	5

Table 2: Confusion matrices for the macro-F1-selected system and the raw-accuracy-best system. The 2B model wins accuracy by turning the task into an almost binary decision and eliminating Ambivalent.

The inability to detect Ambivalent is the central result of the prompting experiments. It is not a small fluctuation. In the final grid, six of the twelve systems predict zero or one Ambivalent example out of thirty. The 2B model predicts zero Ambivalent in three of its four prompt variants. The 0.8B model predicts zero Ambivalent in two variants and only one in the few-shot variant. These systems are not merely underperforming; they frequently fail to represent the middle class as a real decision option.

This is especially damaging because Ambivalent is the conceptual heart of the dataset. A clear reply and a clear non-reply can often be recognized from local cues. Ambivalent requires comparing the question against the response and deciding whether the response is partial, evasive, or only indirectly relevant. Small instruction-following LMs appear to default to surface adequacy: if the answer is topical and fluent, they tend to call it Clear Reply; if it is visibly evasive, they may call it Clear Non-Reply. The in-between category is where they collapse.

5. Comparison Across Homeworks

Homework	System	Evaluation	Acc.	Macro F1	Ambivalent behavior
HW1	TF-IDF + Logistic Regression	official test	~0.63	~0.45	hardest class, but learned useful lexical cues
HW1	GloVe Wiki + Logistic Regression	official test	–	~0.39	mean pooling lost phrase-level evidence
HW1	GloVe Twitter + Logistic Regression	official test	–	~0.38	similar weakness to Wiki embeddings
HW2	BERT-base fine-tuning	official test	0.6299	0.5561	Ambivalent F1 = 0.7216
HW2	DistilBERT fine-tuning	official test	0.5779	0.5066	Ambivalent F1 = 0.6703
HW2	DeBERTa-v3 fine-tuning	official test	0.6688	0.2672	collapsed to all Ambivalent predictions
HW3	Qwen 4B few-shot prompting	balanced validation	0.4667	0.4667	Ambivalent F1 = 0.4000
HW3	Qwen 2B few-shot prompting	balanced validation	0.5000	0.4044	Ambivalent F1 = 0.0000

Table 3: Cross-homework comparison. HW3 is evaluated on a balanced validation sample, so its numbers are diagnostic rather than directly test-comparable. The qualitative comparison is still clear: prompt-only Qwen systems struggle most with Ambivalent.

The comparison yields three conclusions.

First, discriminative training matters. The best BERT model from Homework 2 is the strongest overall system among the completed assignments, with 0.5561 macro F1 on the official test split. It is not merely better on average; it is much better on Ambivalent, where its class F1 is 0.7216. DistilBERT also outperforms the prompt-only systems while using a smaller encoder.

Second, classical TF-IDF remains surprisingly competitive. Its reported macro F1 of about 0.45 is close to the best HW3 prompt result, despite using a much simpler representation. This is not embarrassing for TF-IDF; it is a reminder that the task contains many lexical and phrase-level cues, and that a trained linear classifier can exploit them consistently.

Third, accuracy is unsafe for this task. DeBERTa-v3 in HW2 and Qwen 2B few-shot in HW3 fail in opposite directions. DeBERTa-v3 predicts only Ambivalent; Qwen 2B few-shot predicts no Ambivalent. Both can obtain superficially nontrivial accuracy, but both are invalid as balanced three-class systems. Macro F1 catches both failures.

6. Final Submission System

The final notebook now implements the same policy used in this report:

select by macro F1, break ties by accuracy.

Under this rule, the selected system is `qwen-4b_few-shot`. It is rerun on the full official test split, and the resulting predictions are written to:

`submission_best_prompting_system.csv`.

The selected system is the best available prompting system because it is the only top-performing run that both uses all three labels evenly and maximizes macro F1. The raw-accuracy-best 2B system is deliberately not selected because it has zero Ambivalent recall on validation.

7. Conclusion

The third homework shows the limits of prompt-only classification for this dataset. The pipeline itself works: prompts are generated, Qwen models are run, outputs are

parsed cleanly, all metrics and artifacts are saved, and the final Kaggle submission file is produced with the required name. The failure is not engineering; the failure is classification behavior.

The strongest result is that Ambivalent remains poorly modeled by small and medium Qwen prompting systems. The 4B model partially recovers the class, but only enough to reach 0.4667 macro F1 on the balanced validation sample. Smaller models often erase the class entirely. This is worse than the fine-tuned BERT and DistilBERT systems from Homework 2, and only roughly comparable to the classical TF-IDF baseline from Homework 1. For QEvasion, in-context prompting is convenient and analyzable, but discriminative training is still the more reliable tool.

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